MPSA 2026 - estimation results for paper

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1. Set up

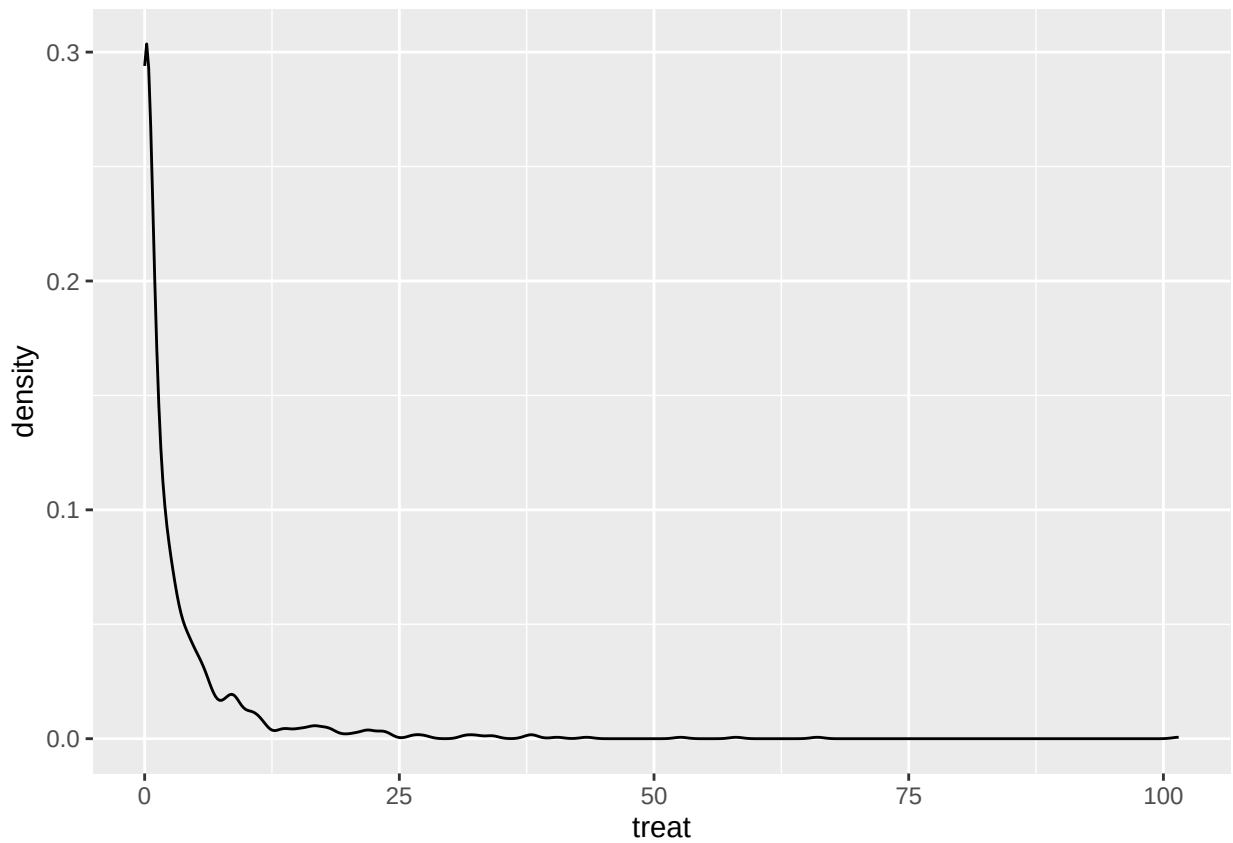
2. Data cleaning

Treatment is disappearances and homicides by the state; calculated as the yearly rate by 10,000

2.1 choosing treatment and covariates

2.2 Covariates

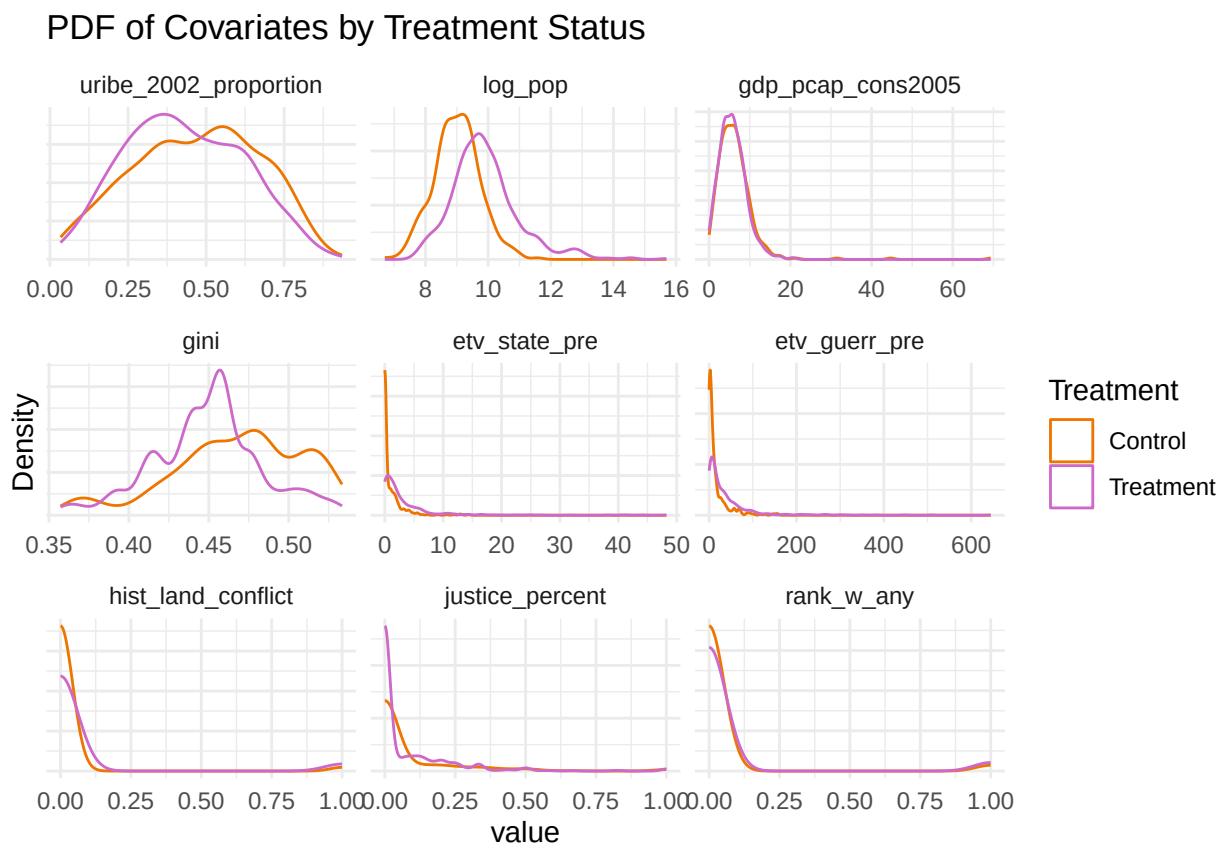
A density plot of the yearly rate of cases per 10,000 per municipality; followed by the exact number and proportion of control municipalities




```
##    0
## 366
```

```
## [1] 0.3638171
```

2.2.1 Covariate balance



	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.430	0.468	0.004	0.001
log_pop	9.899	8.998	0.000	0.000
gdp_pcap_cons2005	5.798	6.399	0.050	0.353
gini	0.449	0.468	0.000	0.000
etv_state_pre	3.142	0.937	0.000	0.000
etv_guerr_pre	37.680	15.411	0.000	0.000
hist_land_conflict	0.069	0.025	0.001	NA
justice_percent	0.076	0.069	0.528	0.000
rank_w_any	0.064	0.038	0.062	0.053

3 Versions of matching and weighting

3.1 Naive regression

3.2 Naive + Control regression

3.3 Matching - covariates

Throws away non-matched control units

- No replacement and one-to-one matching

- with Match(): closest X_i through normalized Euclidean distance
- BiasAdjust = TRUE: bias correction to adjust for multiple continuous variables
- other things to consider: how many matches, which algorithm (eg Coarse, pre-processing), whether to drop treatment units to remove bias, calipers to remove units with no common support

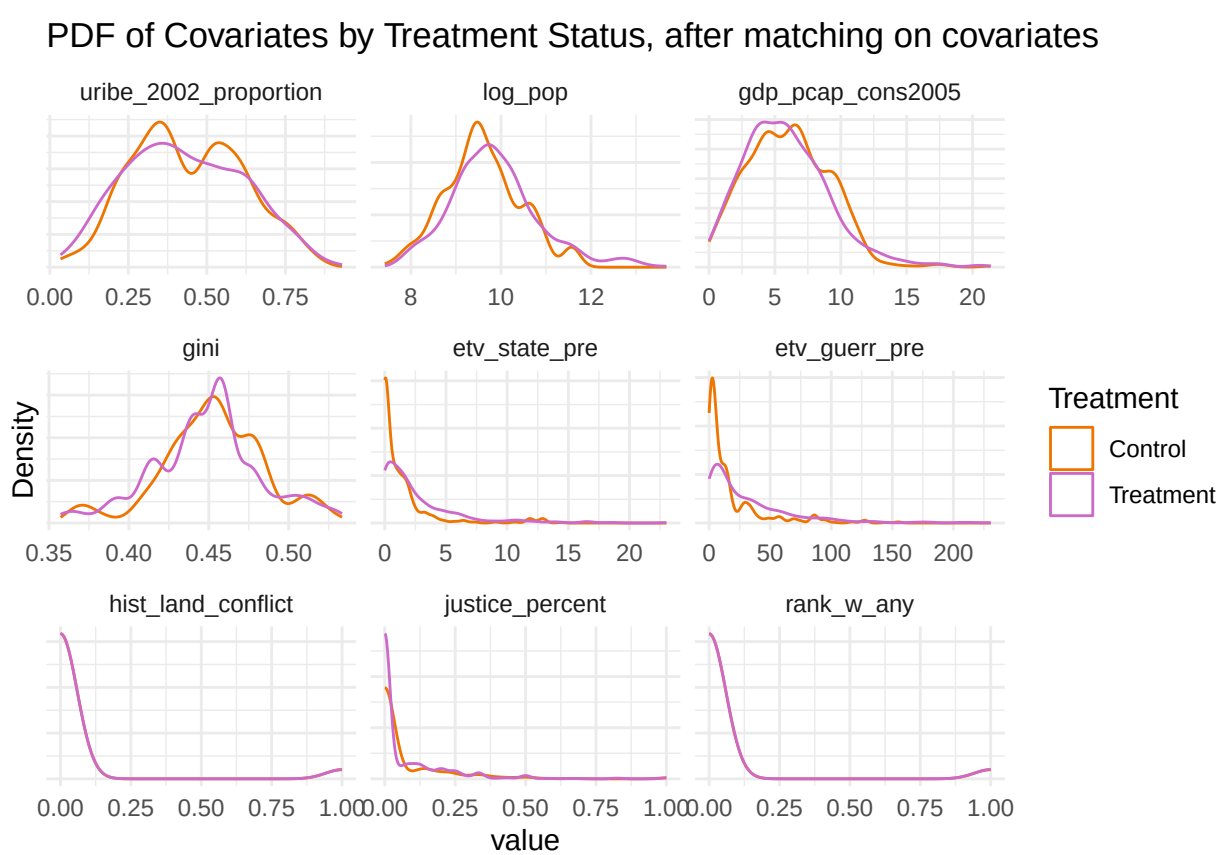
First, creating a matched dataset:

```
##
##   0   1
## 606 606
```

Should we run it multiple times? The results don't change much

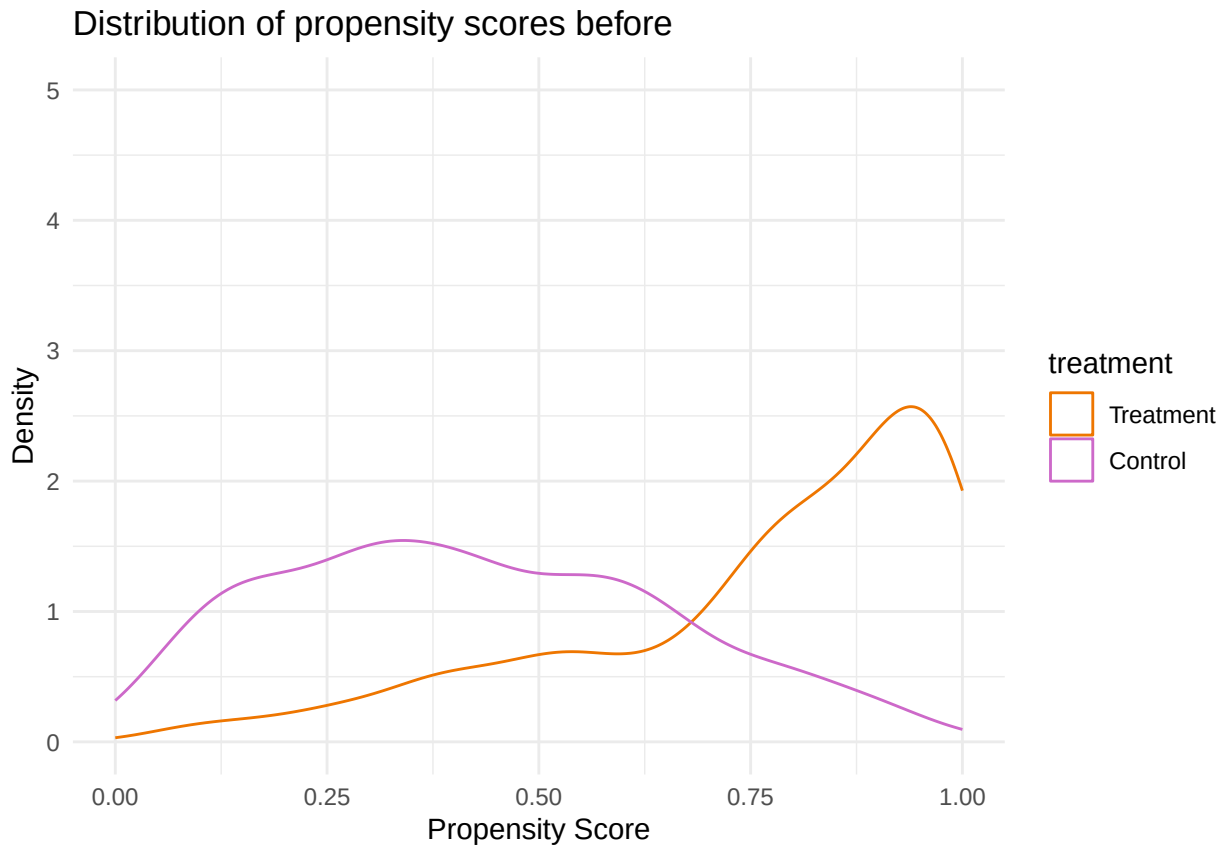
Then, checking the balance after the matching

	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.431	0.442	0.006	0.092
log_pop	9.867	9.591	0.000	0.000
gdp_pcap_cons2005	5.798	5.996	0.018	0.016
gini	0.449	0.453	0.000	0.001
etv_state_pre	2.588	1.354	0.000	0.000
etv_guerr_pre	30.552	17.228	0.000	0.000
hist_land_conflict	0.059	0.059	1.000	NA
justice_percent	0.070	0.059	0.000	0.000
rank_w_any	0.060	0.059	0.101	0.834



Finally, calculating ATT

3.4 Matching - Propensity score matching

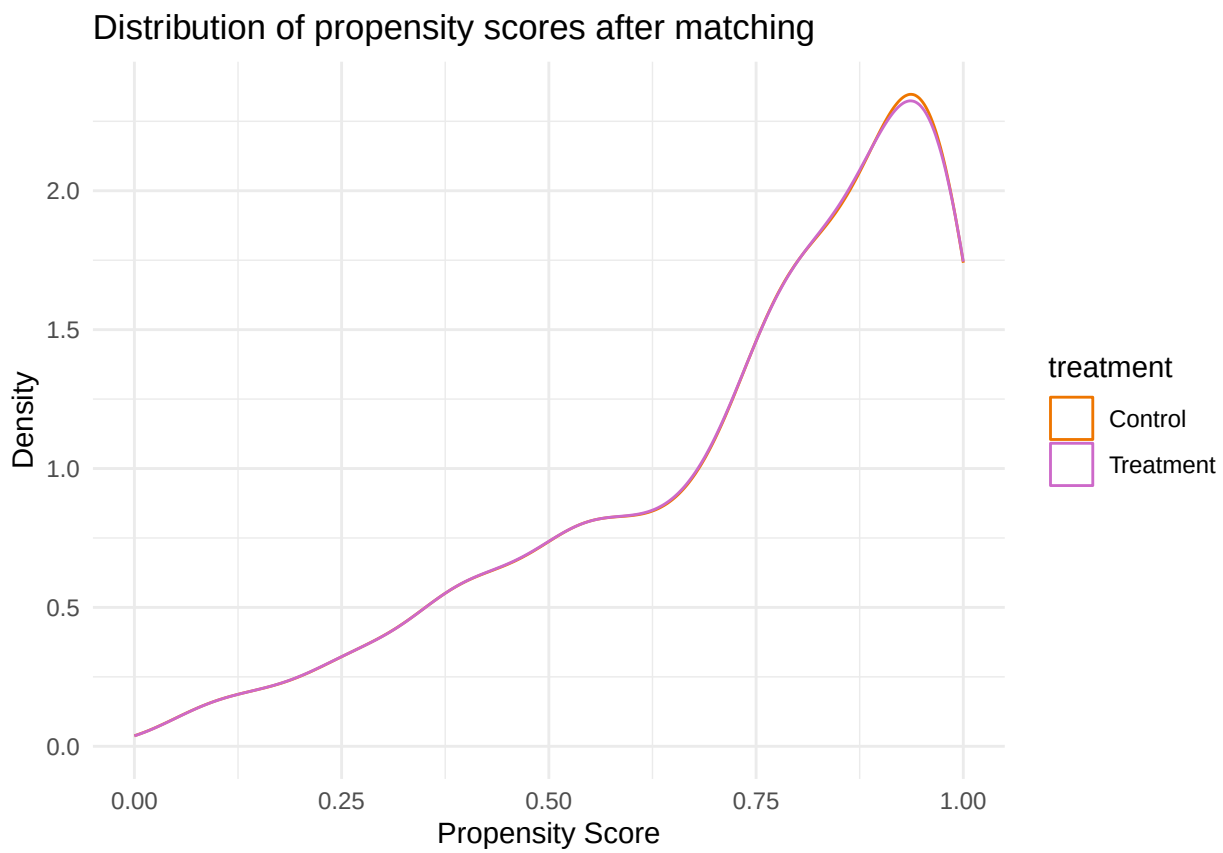


Compare units that have a comparable probability of getting treated to match:

```
##
##   0   1
## 711 711
```

Checking balance after matching:

	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.430	0.512	0.000	0.000
log_pop	9.899	9.532	0.000	0.000
gdp_pcap_cons2005	5.798	6.469	0.000	0.000
gini	0.449	0.454	0.006	0.000
etv_state_pre	3.142	5.248	0.000	0.000
etv_guerr_pre	37.680	32.823	0.044	0.000
hist_land_conflict	0.069	0.034	0.006	NA
justice_percent	0.076	0.051	0.002	0.000
rank_w_any	0.064	0.058	0.643	0.672

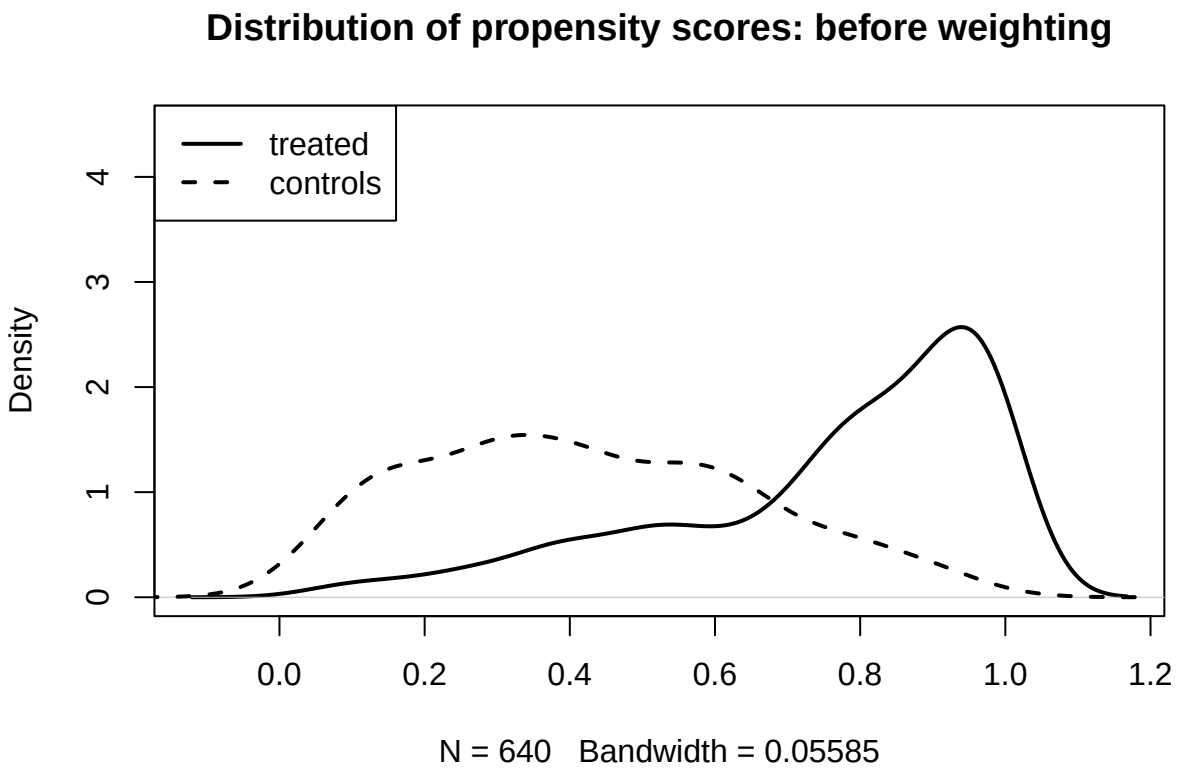


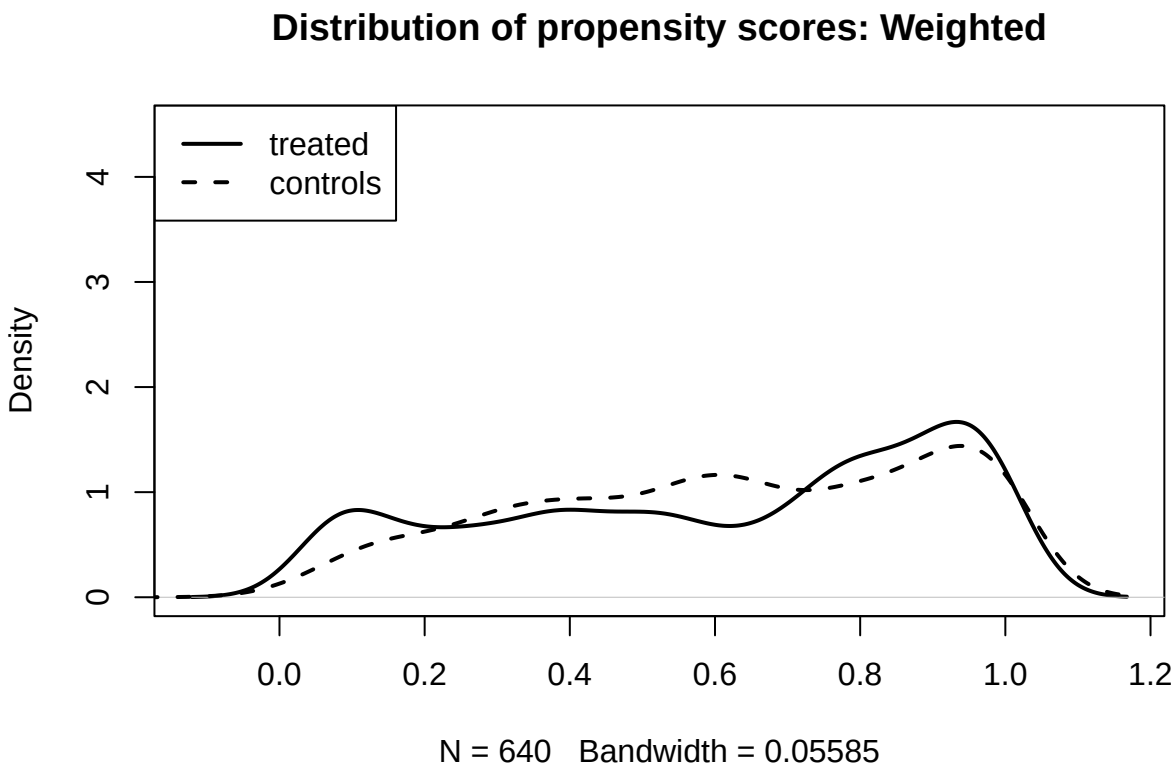
Calculating the ATT:

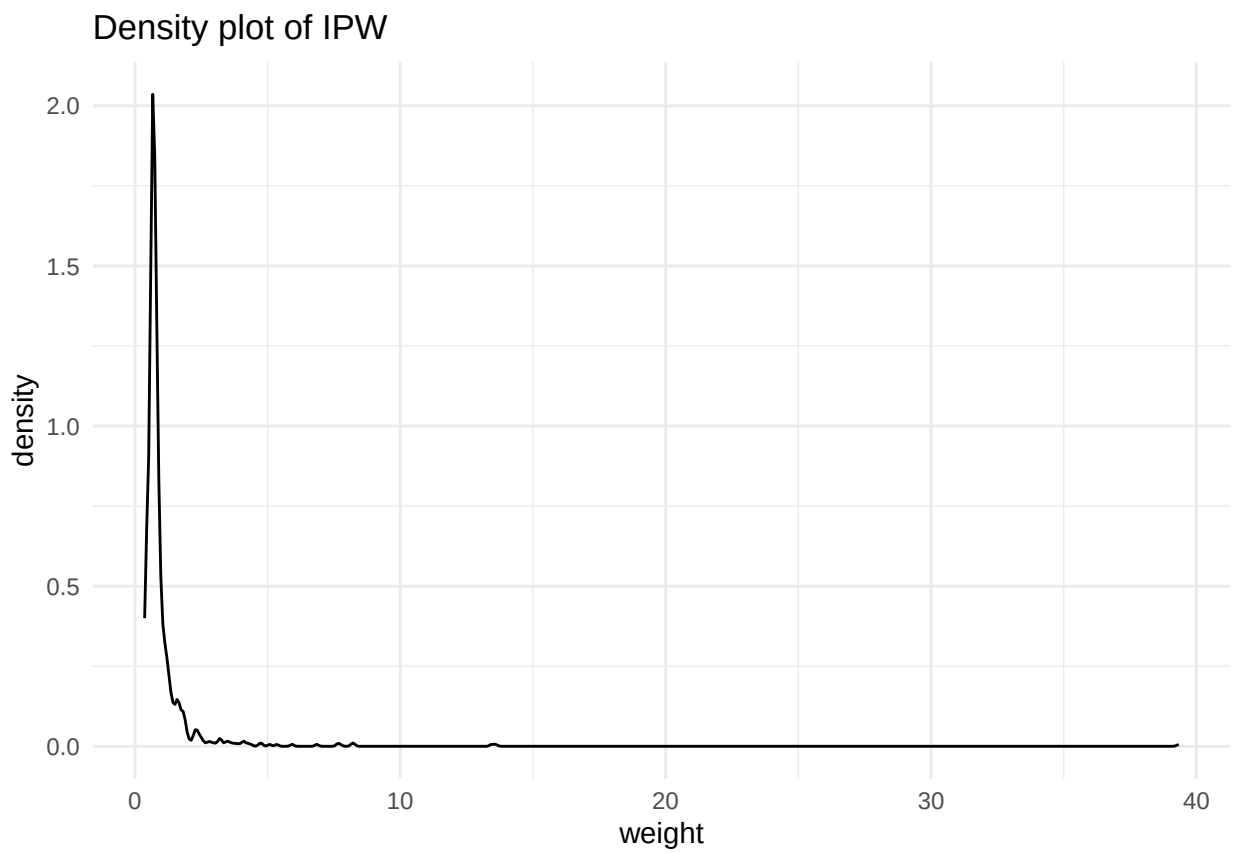
The Match() propensity scores afterwards are very close, but the data has been significantly manipulated: there are many duplicates of both control and treatment

3.5 Weighting - propensity scores

Inverse Probability-of-Treatment Weighting (IPW):


Distribution of propensity scores: before weighting
treated
controls
Density
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
N = 640 Bandwidth = 0.05585


Distribution of propensity scores: Weighted
treated
controls
Density
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
N = 640 Bandwidth = 0.05585

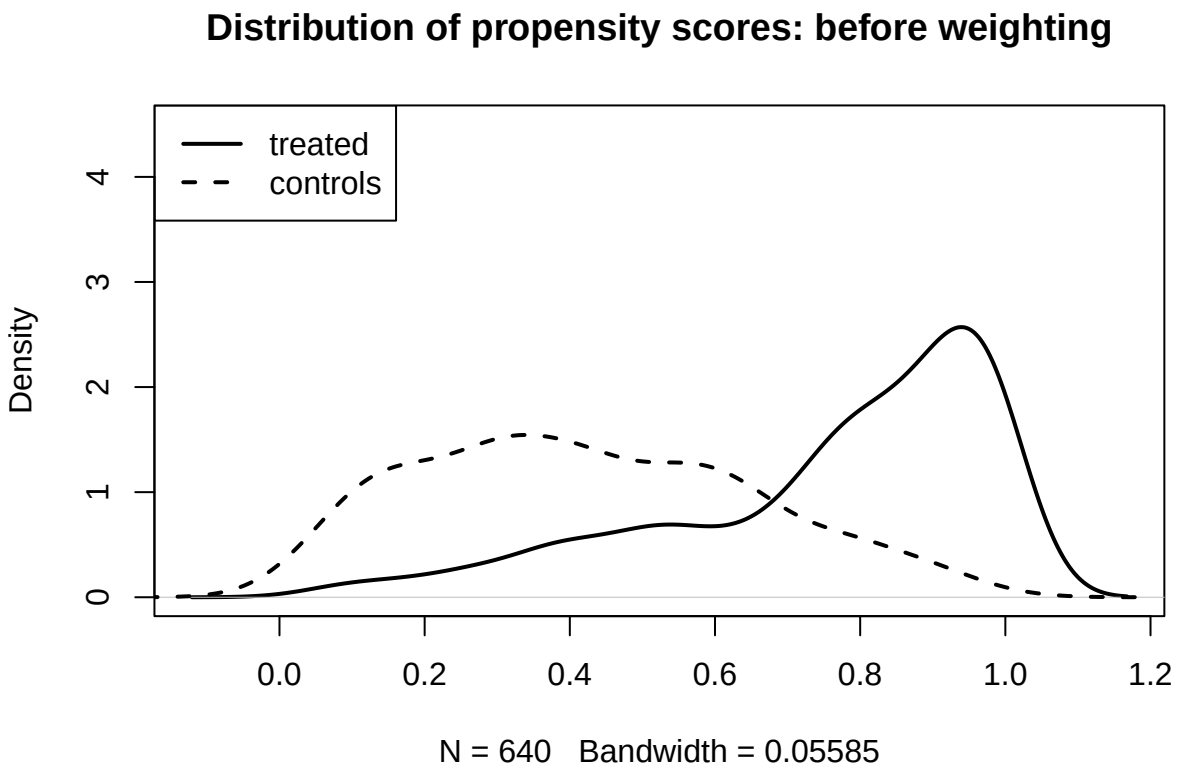


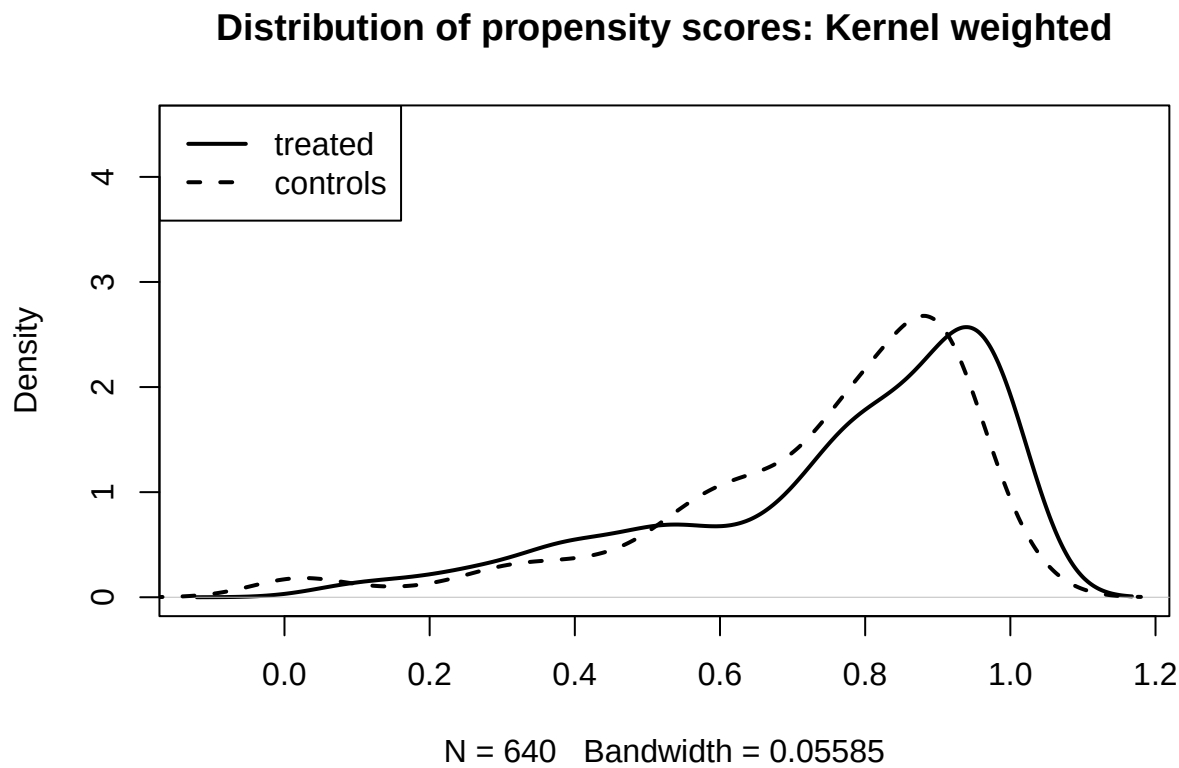
Check how extreme the weights get / how few units are doing most of the work - it means large bias

Take regressions:

3.6 Weighting - Kernel balancing weights

Propensity scores before and after kernel weighting


Distribution of propensity scores: before weighting
treated
controls
Density
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
N = 640 Bandwidth = 0.05585

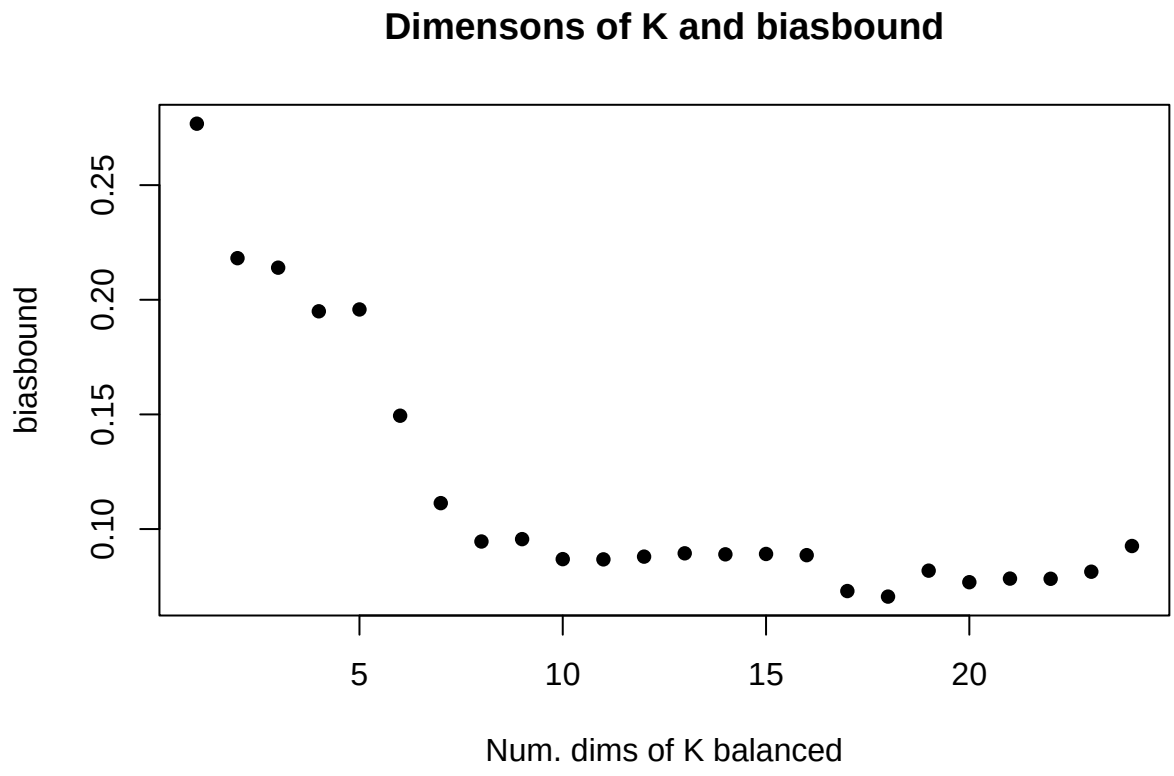


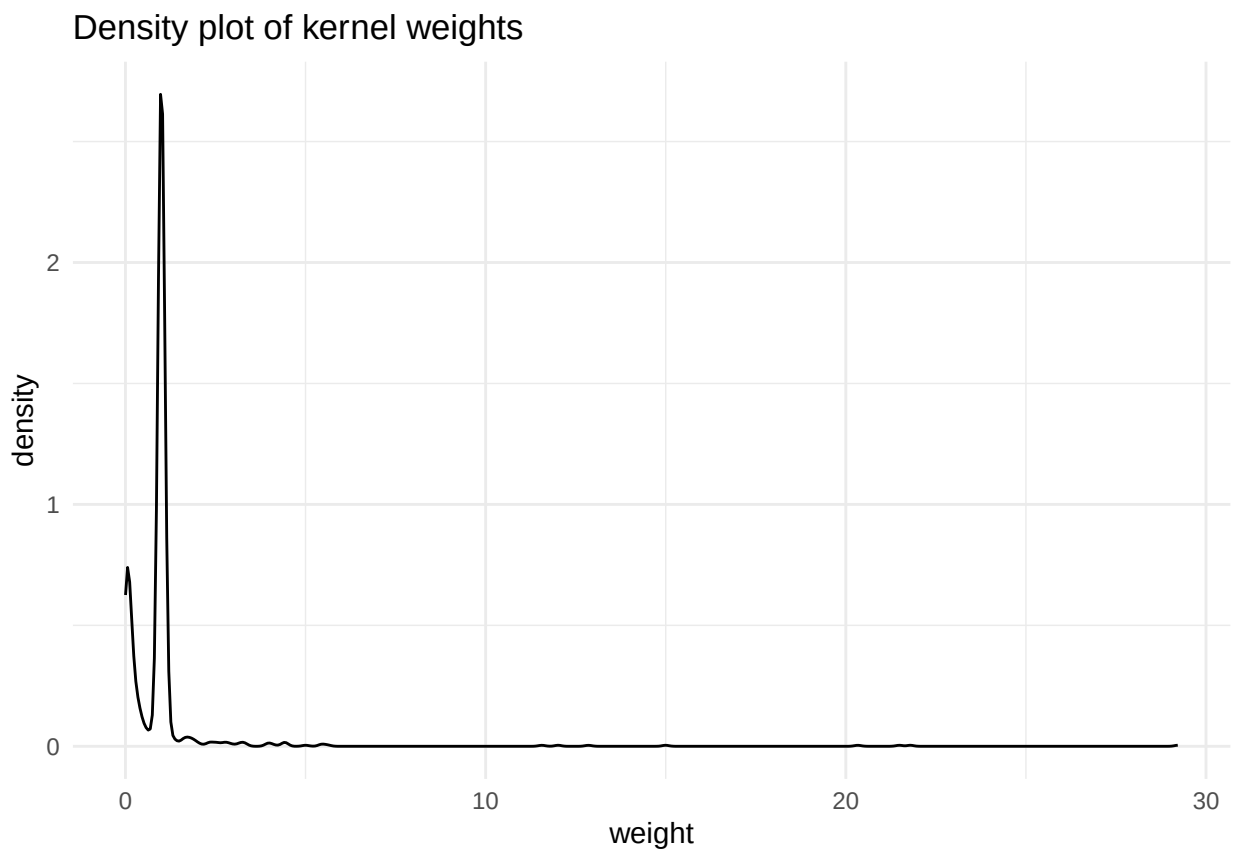
Balance tables, before and after kernel weighting

```
## Balance Measures
##                          Type  M.0.Un  M.1.Un Diff.Un
## uribe_2002_proportion Contin.  0.4677  0.4302 -0.1926
## log_pop               Contin.  8.9984  9.8988  0.9718
## gdp_pcap_cons2005     Contin.  6.3991  5.7980 -0.1367
## gini                  Contin.  0.4677  0.4491 -0.4942
## etv_state_pre         Contin.  0.9374  3.1416  0.5558
## etv_guerr_pre         Contin. 15.4113 37.6798  0.4807
## hist_land_conflict     Binary  0.0246  0.0688  0.0442
## justice_percent       Contin.  0.0688  0.0757  0.0421
## rank_w_any            Contin.  0.0383  0.0643  0.1184
##
## Sample sizes
##     Control Treated
## All     366     640
```

```
## Balance Measures
##                          Type M.0.Adj M.1.Adj Diff.Adj
## uribe_2002_proportion Contin.  0.4384  0.4302  -0.0436
## log_pop               Contin.  9.7159  9.8988   0.1690
## gdp_pcap_cons2005     Contin.  6.7312  5.7980  -0.2860
## gini                  Contin.  0.4501  0.4491  -0.0306
## etv_state_pre         Contin.  2.3720  3.1416   0.1461
## etv_guerr_pre         Contin. 29.4972 37.6798   0.1352
## hist_land_conflict     Binary  0.0637  0.0688   0.0051
```

```
## justice_percent       Contin.  0.0863  0.0757  -0.0699
## rank_w_any            Contin.  0.0591  0.0643   0.0213
##
## Effective sample sizes
##            Control Treated
## Unadjusted  366.       640
## Adjusted     39.54     640
```





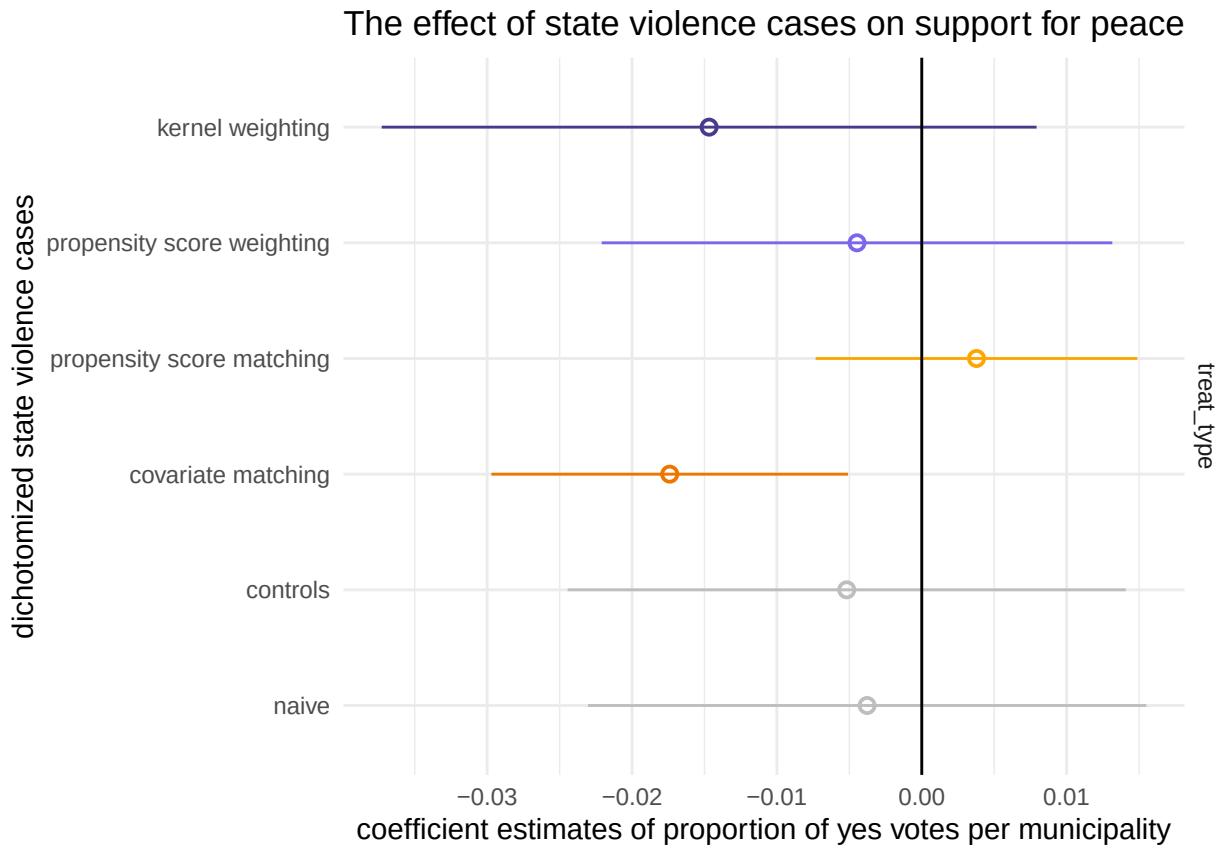
Where treatment and control are the most balanced (ebalance convergence) and bias bound is as low as possible

5. Results

Table 4: The effect of state violence cases on support for peace

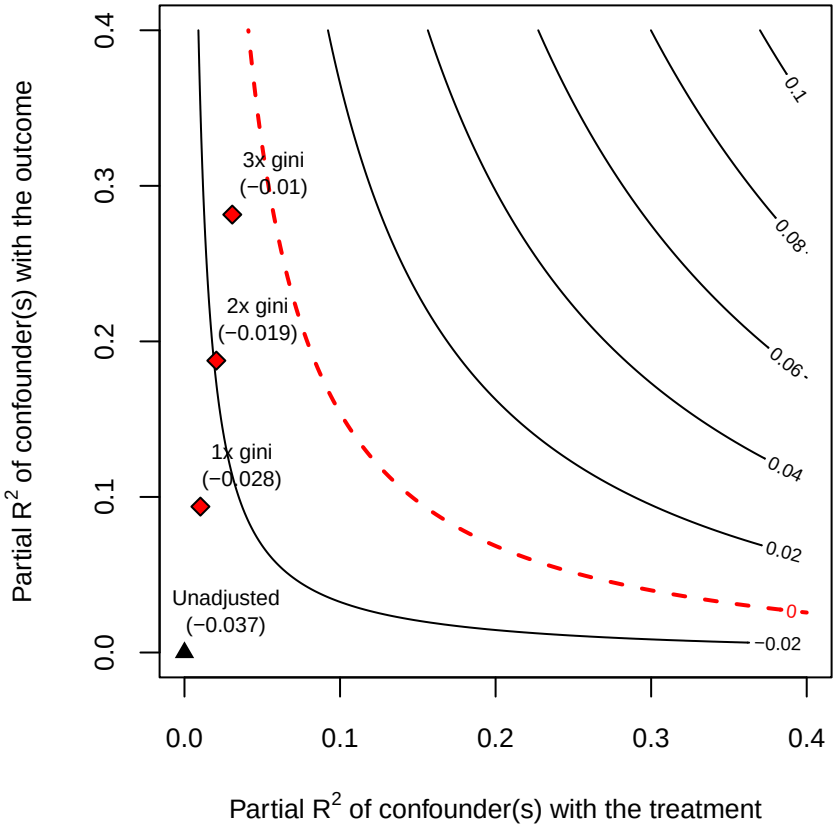
			Matching		Weighting	
	Naive	Controls	Covariates	Propensity scores		Kernel
	(1)	(2)	(3)	(4)	(5)	(6)
(Intercept)	0.496***	0.481***	0.507***	0.466***	0.478***	0.501***
	(0.008)	(0.008)	(0.004)	(0.003)	(0.007)	(0.011)
treatment	−0.004	−0.005	−0.017**	0.004	−0.004	−0.015
	(0.010)	(0.010)	(0.006)	(0.006)	(0.009)	(0.012)
uribe vote		−0.318***	−0.398***	−0.298***	−0.320***	−0.383***
		(0.034)	(0.031)	(0.028)	(0.034)	(0.059)
population		−0.028**	−0.007	−0.019*	−0.023*	−0.023*
		(0.010)	(0.006)	(0.008)	(0.011)	(0.009)
gdp per capita		−0.002	−0.010***	−0.002*	−0.003+	−0.002
		(0.001)	(0.002)	(0.001)	(0.001)	(0.001)
gini		−1.290***	−1.114***	−1.335***	−1.333***	−1.220**
		(0.215)	(0.186)	(0.157)	(0.253)	(0.370)
state violence		−0.006*	−0.007***	−0.010***	−0.009***	−0.011***
		(0.003)	(0.001)	(0.001)	(0.001)	(0.002)
guerilla violence		−0.001***	−0.001***	−0.001***	−0.001*	−0.001***
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
historical land conflict		0.020	0.077***	0.033	0.022	0.003
		(0.031)	(0.012)	(0.021)	(0.033)	(0.035)
judicial capacity		0.019	−0.027	−0.015	−0.001	0.019
		(0.048)	(0.046)	(0.053)	(0.047)	(0.068)
military rank		0.078**	0.020	0.077***	0.057	0.031
		(0.024)	(0.013)	(0.017)	(0.043)	(0.046)
Num.Obs.	1006	1006	1212	1422	1006	1006
R2	0.000	0.450	0.524	0.595	0.497	0.533
R2 Adj.	−0.001	0.439	0.517	0.590	0.488	0.524
RMSE	0.15	0.11			0.11	0.11

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001



6. Sensitivity analysis

Which covariate as a benchmark?



Sensitivity with regular lm() only; ipw/matching weights mess it up – how to add other versions?